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Classical Field Theory

In what follows we will consider rather general field theories. The only guiding principles that we will use in constructing these theories are (a) symmetries and (b) a generalized Least Action Principle.

2.1 Relativistic Invariance

Before we saw three examples of relativistic wave equations. They are the Maxwell equations for classical electromagnetism, the Klein-Gordon equation and the Dirac equation. Maxwell's equations govern the dynamics of a *vector* field, the vector potentials $A^\mu(x) = (A^0(x), \boldsymbol{A}(x))$, whereas the Klein-Gordon equation describes excitations of a *scalar* field $\phi(x)$ and the Dirac equation governs the behavior of the four-component spinor field $\psi_\alpha(x)(\alpha = 0, 1, 2, 3)$. Each one of these fields transforms in a very definite way under the group of Lorentz transformations, the Lorentz group. The Lorentz group is defined as a group of linear transformations Λ of Minkowski space-time $\mathcal{M}$ onto itself $\Lambda : \mathcal{M} \mapsto \mathcal{M}$ such that the new coordinates are related to the old ones by a linear (Lorentz) transformation

$$x'^\mu = \Lambda^\mu_\nu x^\nu \tag{2.1}$$

The space-time components of Λ^0_i are the *Lorentz boosts* which relate inertial reference frames moving at relative velocity $\vec{v}$ from each other. Thus,

2.6 Classical Electromagnetism as a field theory

We now turn to the problem of the electromagnetic field generated by a set of sources. Let $\rho(x)$ and $\boldsymbol{j}(x)$ represent the charge density and current at a point x of space-time. Charge conservation requires that a continuity equation has to be obeyed.

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{j} = 0 \tag{2.151}$$

Given an initial condition, *i.e.,* the values of the electric field $\boldsymbol{E}(x)$ and the magnetic field $\boldsymbol{B}(x)$ at some t_0 in the past, the time evolution is governed by Maxwell's equations

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \rho \qquad\qquad \boldsymbol{\nabla} \cdot \boldsymbol{B} = 0 \tag{2.152}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B} - \frac{1}{c}\frac{\partial \boldsymbol{E}}{\partial t} = \boldsymbol{j} \qquad\qquad \boldsymbol{\nabla} \times \boldsymbol{E} + \frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t} = 0 \tag{2.153}$$

It is possible to recast these statements in a manner in which (a) the relativistic covariance is apparent and (b) the equations follow from a Least Action Principle. A convenient way to see the above is to consider the electromagnetic field tensor $F^{\mu\nu}$ which is the (contravariant) antisymmetric real tensor

$$F^{\mu\nu} = -F^{\nu\mu} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & -B^3 & B^2 \\ E^2 & B^3 & 0 & -B^1 \\ E^3 & -B^2 & B^1 & 0 \end{pmatrix} \tag{2.154}$$

In other words

$$\begin{aligned} F^{0i} &= -F^{i0} = -E^i \\ F^{ij} &= -F^{ji} = \epsilon^{ijk} B^k \end{aligned} \tag{2.155}$$

where ϵ^{ijk} is the third-rank Levi-Civita tensor:

$$\epsilon^{ijk} = \begin{cases} 1 & \text{if } (ijk) \text{ is an even permutation of } (123) \\ -1 & \text{if } (ijk) \text{ is an odd permutation of } (123) \\ 0 & \text{otherwise} \end{cases} \tag{2.156}$$

The dual tensor $\tilde{F}_{\mu\nu}$ is defined by

$$\tilde{F}^{\mu\nu} = -\tilde{F}^{\nu\mu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \tag{2.157}$$

where $\epsilon^{\mu\nu\rho\sigma}$ is the fourth rank Levi-Civita tensor. In particular

$$\tilde{F}^{\mu\nu} = \begin{pmatrix} 0 & -B^1 & -B^2 & -B^3 \\ B^1 & 0 & E^3 & -E^2 \\ B^2 & -E^3 & 0 & E^2 \\ B^3 & E^2 & -E^2 & 0 \end{pmatrix} \tag{2.158}$$

With these notations, we can rewrite Maxwell's equations in the manifestly covariant form

$$\partial_\mu F^{\mu\nu} = j^\nu \qquad \text{(Equation of Motion)} \tag{2.159}$$

$$\partial_\mu \tilde{F}^{\mu\nu} = 0 \qquad \text{(Bianchi Identity)} \tag{2.160}$$

$$\partial_\mu j^\mu = 0 \qquad \text{(Continuity Equation)} \tag{2.161}$$

By inspection we see that the field tensor $F^{\mu\nu}$ and the dual field tensor $\widetilde{F}^{\mu\nu}$ map into each other by exchanging the electric and magnetic fields with each others. This *electro-magnetic duality* would be an exact property of electrodynamics if in addition to the electric charge current j^μ the Bianchi Identity Eq.(2.160) included a magnetic charge current (of magnetic monopoles).

At this point it is convenient to introduce the vector potential A^μ whose contravariant components are

$$A^\mu(x) = \left(\frac{A^0}{c}, \boldsymbol{A}\right) \tag{2.162}$$

The current 4-vector $j^\mu(x)$

$$j^\mu(x) = (\rho c, \boldsymbol{j}) \equiv (j^0, \boldsymbol{j}) \tag{2.163}$$

The electric field strength $\boldsymbol{E}$ and the magnetic field $\boldsymbol{B}$ are defined to be

$$\begin{aligned} \boldsymbol{E} &= -\frac{1}{c}\boldsymbol{\nabla} A^0 - \frac{1}{c}\frac{\partial \boldsymbol{A}}{\partial t} \\ \boldsymbol{B} &= \boldsymbol{\nabla} \times \boldsymbol{A} \end{aligned} \tag{2.164}$$

In a more compact, relativistically covariant, notation we write

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu \tag{2.165}$$

In terms of the vector potential A^μ, Maxwell's equations have the following additional *local* (or *gauge*) symmetry

$$A^\mu(x) \mapsto A^\mu(x) + \partial^\mu \Phi(x) \tag{2.166}$$

where $\Phi(x)$ is an *arbitrary smooth* function of the space-time coordinates x^μ. It is easy to check that, under the transformation of Eq.(2.166), the field

strength remains *invariant i.e.,* $F^{\mu\nu} \mapsto F^{\mu\nu}$. This property is called *Gauge Invariance* and it plays a *fundamental* role in modern physics.

By directly substituting the definitions of the magnetic field $\boldsymbol{B}$ and the electric field $\boldsymbol{E}$ in terms of the 4-vector A^μ into Maxwell's equations, we obtain the wave equation. Indeed, the equation of motion

$$\partial_\mu F^{\mu\nu} = j^\nu \tag{2.167}$$

yields the equation for the vector potential

$$\partial^2 A^\nu - \partial^\nu(\partial_\mu A^\mu) = j^\nu \tag{2.168}$$

This is the *wave equation.*

We can now use the gauge invariance to further restrict the vector potential A^μ, without which A^μ is not completely determined. These restrictions are known as the procedure of *fixing a gauge.* The choice

$$\partial_\mu A^\mu = 0 \tag{2.169}$$

known as the Lorentz gauge, yields the simpler (and standard) wave equation

$$\partial^2 A^\mu = j^\mu \tag{2.170}$$

Notice that the Lorentz gauge preserves Lorentz covariance.

Another "popular" choice is the radiation (or Coulomb) gauge

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0 \tag{2.171}$$

which yields (in units with $c = 1$)

$$\partial^2 A^\nu - \partial^\nu(\partial_0 A^0) = j^\nu \tag{2.172}$$

which is not Lorentz covariant. In the absence of external sources, $j^\nu = 0$, we can further make the choice $A^0 = 0$. This choice reduces the set of three equations, one for each spacial component of $\boldsymbol{A}$, which satisfy

$$\partial^2 \boldsymbol{A} = 0, \quad \text{provided} \quad \boldsymbol{\nabla} \cdot \boldsymbol{A} = 0 \tag{2.173}$$

The solutions are, as we well know, plane waves of the form

$$\boldsymbol{A}(x) = \boldsymbol{A}\, e^{i(p_0 x_0 - \boldsymbol{p}\cdot\boldsymbol{x})} \tag{2.174}$$

which are only consistent if $p_0^2 - \boldsymbol{p}^2 = 0$ and $\boldsymbol{p} \cdot \boldsymbol{A} = 0$. This choice is also known as the *transverse* gauge.

We can also regard the electromagnetic field as a dynamical system and

construct a Lagrangian picture for it. Since the Maxwell equations are local, gauge invariant and Lorentz covariant, we should demand that the Lagrangian density should be local, gauge invariant and Lorentz invariant. A simple choice is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - j_\mu A^\mu \tag{2.175}$$

This Lagrangian density is manifestly Lorentz invariant. Gauge invariance is satisfied if and only if j_μ is a conserved current, i.e. if $\partial_\mu j^\mu = 0$, since under a gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \Phi(x)$ the field strength does not change but the source term does. Hence,

$$\begin{aligned}\int d^4x\; j_\mu A^\mu \mapsto & \int d^4x \left[j_\mu A^\mu + j_\mu \partial^\mu \Phi\right] \\ &= \int d^4x\; j_\mu A^\mu + \int d^4x\; \partial^\mu (j_\mu \Phi) - \int d^4x\; \partial^\mu j_\mu\; \Phi \end{aligned} \tag{2.176}$$

If the sources vanish at infinity, $\lim_{|x|\to\infty} j_\mu = 0$, the surface term can be dropped. Thus the action $S = \int d^4x \mathcal{L}$ is gauge-invariant if and only if the current j^μ is locally conserved,

$$\partial_\mu j^\mu = 0 \tag{2.177}$$

We can now derive the equations of motion by demanding that the action S be stationary, *i.e.,*

$$\delta S = \int d^4x \left[\frac{\delta \mathcal{L}}{\delta A^\mu}\delta A^\mu + \frac{\delta \mathcal{L}}{\delta \partial^\mu A^\mu}\delta \partial^\mu A^\mu\right] = 0 \tag{2.178}$$

Once again, we can integrate by parts to get

$$\delta S = \int d^4x\; \partial^\nu \left[\frac{\partial \mathcal{L}}{\delta \partial^\mu A^\mu}\delta A^\mu\right] + \int d^4x\; \delta A^\mu \left[\frac{\delta \mathcal{L}}{\delta A^\mu} - \partial^\nu \left(\frac{\delta \mathcal{L}}{\delta \partial^\nu A^\mu}\right)\right] \tag{2.179}$$

If we demand that at the surface $\delta A^\mu = 0$, we get

$$\frac{\delta \mathcal{L}}{\delta A^\mu} = \partial^\nu \left(\frac{\delta \mathcal{L}}{\delta \partial^\nu A^\mu}\right) \tag{2.180}$$

Explicitly, we find

$$\frac{\delta \mathcal{L}}{\delta A^\mu} = -j_\mu, \quad \text{and} \quad \frac{\delta \mathcal{L}}{\delta \partial^\nu A^\mu} = F^{\mu\nu} \tag{2.181}$$

Thus, we obtain

$$j^\mu = -\partial_\nu F^{\mu\nu} \tag{2.182}$$

or, equivalently

$$j^\nu = \partial_\mu F^{\mu\nu} \tag{2.183}$$